Advanced Statistical Mechanics
Assignment 01

Sagnik Seth 22MS026

Question 1 (Yang-Lee edge):

Consider the following grand partition function for a hypothetical fluid

$$\Xi(z, V) = (1 + z)^{[\alpha V]}(1 + z^{[\alpha V]}).$$

Here $[x]$ denotes the integer part of x.

(a) Determine the zeros of $\Xi(z, V)$ and (roughly) plot them on the complex z plane. Here z denotes the fugacity, $z = e^{\beta\mu}$.

(b) Show that the zeros of $\Xi(z, V)$ in the thermodynamic limit hit the real z axis at $z = 1$.

(c) Evaluate

$$\Theta = \lim_{V\to\infty} \frac{1}{V} \ln \Xi(z, V)$$

for $z < 1$ and $z > 1$ and show that it is continuous but non-analytic.

(d) Given that the particle density is $n = z\frac{\partial\Theta}{\partial z}$, show that there is a density jump at the phase transition.

(e) Using the fact $\beta p = \Theta$, derive the equations of state $p(n)$, for $z < 1$ and $z > 1$ by eliminating z from the expressions of $p(z)$ and $n(z)$.

(a) The roots of the grand partition will satisfy,

$$(z + 1)^{[\alpha V]} = 0 \qquad \text{or} \qquad z^{[\alpha V]} = -1$$

The first condition gives the real root $z = -1$ with multiplicity $[\alpha V]$. The second condition gives $z = \exp\left[\frac{i\pi(2n+1)}{[\alpha V]}\right]$ for $n = 0, 1 \ldots ([\alpha V] - 1)$. The roots lie on the unit circle in the complex plane and the arguments are,

$$\theta_n = \pi\frac{(2n + 1)}{[\alpha V]}$$

.

(b) For $n = 0$, we have $z_0 = e^{i\pi/[\alpha V]}$ which in the thermodynamic limit $V \to \infty$ becomes, $z_0 = 1$. Hence, in the thermodynamic limit, one of the roots of Ξ hits the real axis at $z = 1$

Question 2 (Critical exponents in the mean field Ising model.):

Familiarize yourself with the definition of the critical exponents β, γ, and δ. Show that the specific heat for the mean field Ising model satisfies

$$\begin{cases} c = \dfrac{3Nk_B}{2}, & T < T_c, \\ c = 0, & T > T_c. \end{cases}$$

If the specific heat critical exponent α is defined via the relation

$$c \sim \left| \frac{T - T_c}{T_c} \right|^{-\alpha},$$

what value of α do we infer from the above analysis?

Answer. The partition function of the mean-field Ising model can be written as,

$$\mathcal{Z} = [2\cosh(\beta J\gamma m + \beta h)]^N e^{-\beta\varepsilon_0}$$

Now, the specific heat is defined as $c = \beta^2 \frac{\partial^2}{\partial\beta^2}(\ln \mathcal{Z})$ which gives us,

$$\begin{aligned} c &= \beta^2 \frac{\partial^2}{\partial\beta^2}(\beta\varepsilon_0 + N\ln 2 + N\ln\cosh(\beta J\gamma m + \beta h)) \\ &= N(J\gamma m + h)\beta^2 \frac{\partial}{\partial\beta}(\tanh[\beta J\gamma m + \beta h]) \\ &= N(J\gamma m + h)^2 \big(\mathrm{sech}^2[\beta J\gamma m + \beta h]\big) \end{aligned}$$

Question 3 (q-state Potts model):

A generalization of the Ising model is the q-state Potts model, where one assume that the spin variable at a site can now take q different values, $\sigma_i \in \{1, 2, 3, .., q\}$ and the Hamiltonian is given by,

$$\mathcal{H}_{\text{Potts}} = -J \sum_{\langle ij \rangle} \delta_{\sigma_i\sigma_j} - h \sum_i \delta_{\sigma_i,1}$$

where $\langle ij \rangle$ denotes nearest neighbour interaction. By defining a suitable transformation between the Ising spin variable σ_i and the Kronecker delta $\delta_{\sigma_i\sigma_j}$, show that the for $q = 2$, this Hamiltonian is equivalent to the Ising Hamiltonian.

Answer. The interaction term given here is $\delta_{\sigma_i\sigma_j}$ while in the Ising Hamiltonian, the interaction term is S_iS_j. If both Ising spins are different, the interaction is -1 while for the $q = 2$ Potts model, it is 0. If both Ising spins are same, the interaction is $+1$ which is same as the $q = 2$ Potts model. Thus we can have a linearly transformation, such that

$$\delta_{\sigma_i\sigma_j} = aS_iS_j + b$$

The conditions to satisfy are,

$$0 = b - a \qquad 1 = b + a$$

from which we obtain $a = b = 1/2$. Thus, the transformation is,

$$\delta_{\sigma_i\sigma_j} = \frac{1}{2}(S_iS_j + 1)$$

With this transformation, the Potts Hamiltonian becomes,

$$\begin{aligned}
\mathcal{H}_{\text{Potts}} &= -\frac{J}{2}\sum_{\langle ij\rangle}(S_iS_j+1) - \frac{h}{2}\sum_i(S_i+1) \\
&= -\frac{J}{2}\sum_{\langle ij\rangle}S_iS_j - \frac{h}{2}\sum_i S_i - \underbrace{\frac{J}{2}\sum_{\langle ij\rangle}1 - \frac{h}{2}\sum_i 1}_{\text{const.}} \\
&= -J'\sum_{\langle ij\rangle}S_iS_j - h'\sum_i S_i + \text{const.} \\
&= \mathcal{H}_{\text{Ising}} + \text{const.}
\end{aligned} \tag{1}$$

where $J' \equiv {}^{J}\!/_{2}$ and $h' \equiv {}^{h}\!/_{2}$. Since a constant in the Hamiltonian can be neglected (will cause only a shift in the energy), we obtain that for $q = 2$, the Potts model reduces to the Ising Hamiltonian.

Question 4 (Antiferromagnetic Ising Model):

Consider the Ising model on a two-dimensional square lattice with nearest neighbour interaction but with the interaction term having opposite sign such that antiparallel allignment is favoured.

$$\mathcal{H} = J\sum_{\langle ij\rangle}\sigma_i\sigma_j - h\sum_i\sigma_i \qquad J > 0$$

(a) Show that for $h = 0$, the partition function and hence the free energy is the same as that for the usual Ising model.

(b) Solve the problem in the mean field approximation with $h = 0$.